SIMATS ENGINEERING

CO2 – DRY RUN EVALUATION (ASSESSMENT TOOL 3)

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Course: Artificial Intelligence **Course Code:** CSA17

Question 1 – Breadth-First Search (BFS)

Assumption: the graph is traversed from left to right, with children of B ordered D then E and children of C ordered F then G.

Iteration	Queue before expansion	Expanded node	Queue after expansion	Visited order
1	A	A	B, C	A
2	B, C	B	C, D, E	A, B
3	C, D, E	C	D, E, F, G	A, B, C
4	D, E, F, G	D	E, F, G	A, B, C, D
5	E, F, G	E	F, G	A, B, C, D, E
6	F, G	F	G	A, B, C, D, E, F
7	G	G	Empty	A, B, C, D, E, F, G

1. BFS traversal order: A → B → C → D → E → F → G.

2. Queue contents: After each iteration: [B, C], [C, D, E], [D, E, F, G], [E, F, G], [F, G], [G], and [].

Question 2 – Depth-First Search (DFS)

Using left-to-right order, DFS follows one branch as deeply as possible before backtracking.

Step	Action	Stack (top at right)	Visited order
1	Visit A; push C, B	[C, B]	A
2	Visit B; push E, D	[C, E, D]	A, B
3	Visit D	[C, E]	A, B, D
4	Backtrack; visit E	[C]	A, B, D, E
5	Backtrack; visit C; push G, F	[G, F]	A, B, D, E, C
6	Visit F	[G]	A, B, D, E, C, F
7	Visit G	[]	A, B, D, E, C, F, G

DFS traversal order: A → B → D → E → C → F → G.

Note: The source sheet shows the DFS instruction but no visible table headings/rows. The sequence above is a clear dry-run format using the given graph.

Question 3 – 8-Puzzle Using BFS

Initial state

```
1   3   6
5   0   2
4   7   8
```

Goal state

```
1   2   3
4   5   6
7   8   0
```

Important note: a complete BFS trace for the 8-puzzle can contain many states. The five-row table below is a compact sample trace, not a claim that the goal is reached in five expansions.

Iteration	Current state (0 is blank)	Queue before expansion	Queue after expansion (new states, in move order U, D, L, R)
1	136 / 502 / 478	[Initial]	136/052/478; 106/532/478; 136/520/478; 136/502/478 (skip duplicate)
2	136 / 052 / 478	States generated in iteration 1	036/152/478; 136/452/078; 106/532/478 (duplicate skipped); 136/502/478 (duplicate skipped)
3	106 / 532 / 478	Remaining frontier states	016/532/478; 160/532/478; 126/530/478
4	136 / 520 / 478	Remaining frontier states	130/526/478; 136/528/470
5	036 / 152 / 478	Remaining frontier states	Blank moves generate its legal neighbouring states; repeated states are skipped.

1. Which node is expanded in each iteration? The queue is expanded in FIFO order. In the sample table: the initial state, then 136/052/478, then 106/532/478, then 136/520/478, then 036/152/478.

2. How many states are generated in total? The exact total depends on continuing BFS until the goal is reached and on duplicate-state handling. It cannot be determined from only the five sample rows; continue expanding the FIFO queue until 123/456/780 is dequeued.

3. Complete solution path: A complete path is not provided here because the assignment asks for BFS to be traced until the goal is reached; the five-row table is only a sample and does not reach the goal. Continue BFS with visited-state tracking to obtain the full shortest path.

4. Why does BFS find the shortest solution? In an unweighted state space, every move has equal cost. BFS explores all states at depth 0, then depth 1, then depth 2, and so on. Therefore, the first time it reaches the goal, the path uses the minimum number of moves.

Code of Conduct

I, A.K. Kishore Kumar (Reg. No. 1913260079), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

Signature of the student: _____